

Guide Replication Transactionnelle SQL Server 2022 sur Ubuntu 24.04

Ce document fournit les commandes bash et scripts T-SQL pour configurer une replication transactionnelle entre deux instances SQL Server 2022 sur Ubuntu 24.04 (Serveur A = Publieur, Serveur B = Abonne). Toutes les commandes sont pretes a l'emploi.

1 Prérequis système (bash)

1.1 Installer et activer SQL Server Agent (A et B)

```
sudo apt-get update
sudo apt-get install -y mssql-server-agent
sudo systemctl enable --now mssql-server-agent
sudo systemctl restart mssql-server
sudo systemctl status mssql-server-agent --no-pager
```

1.2 Ouvrir le port 1433 dans UFW (A et B)

```
sudo ufw allow 1433/tcp
sudo ufw reload
sudo ufw status
```

1.3 Resolution de noms via /etc/hosts (adapter IPs, A et B)

```
sudo sh -c 'cat >> /etc/hosts <<EOF
10.0.0.10 publieur
10.0.0.11 abonne
EOF'
```

Verifier :

```
ping -c 2 publieur
ping -c 2 abonne
```

1.4 Dossier snapshot (Publieur)

```
sudo mkdir -p /var/opt/mssql/data/ReplData
sudo chown mssql:mssql /var/opt/mssql/data/ReplData
sudo chmod 770 /var/opt/mssql/data/ReplData
```

2 Configuration SQL (T-SQL)

Executer dans sqlcmd, ADS ou SSMS.

2.1 Variables a adapter

```
DECLARE @DistDb sysname = N'distribution';
DECLARE @Publisher sysname = N'publieur';
DECLARE @Subscriber sysname = N'abonne';
DECLARE @PublicationDb sysname = N'TaBaseSource';
DECLARE @PublicationName sysname = N'TaPublication';
DECLARE @ArticleTable sysname = N'dbo.TaTableARepl';
DECLARE @SnapshotDir nvarchar(260) = N'/var/opt/mssql/data/ReplData';
DECLARE @ReplLogin sysname = N'repl_user';
DECLARE @ReplPwd nvarchar(128) = N'TonMDPSecurise!23';
```

2.2 Compte SQL repleur (Publieur)

```
USE master;
IF NOT EXISTS (SELECT 1 FROM sys.sql_logins WHERE name = @ReplLogin)
BEGIN
    CREATE LOGIN [repl_user] WITH PASSWORD = @ReplPwd, CHECK_POLICY = ON,
        CHECK_EXPIRATION = OFF;
END;
EXEC sp_addsrvrolemember @loginame = @ReplLogin, @rolename = N'sysadmin';
```

2.3 Configurer le Distributeur (Publieur)

```
-- Activer la distribution locale
EXEC sp_adddistributor @distributor = @Publisher, @password = N'StrongDistPwd!23';

-- Creer la DB distribution
EXEC sp_adddistributiondb
    @database = @DistDb,
    @data_folder = N'/var/opt/mssql/data',
    @log_folder = N'/var/opt/mssql/data',
    @log_file_size = 2, @min_distretention = 0, @max_distretention = 72,
    @history_retention = 48;

-- Lier le distributeur au publieur
EXEC sp_adddistpublisher
    @publisher = @Publisher,
    @distribution_db = @DistDb,
    @security_mode = 0,
    @login = @ReplLogin,
    @password = @ReplPwd,
    @working_directory = @SnapshotDir,
    @trusted = N'false',
    @thirdparty_flag = 0,
    @publisher_type = N'MSSQLSERVER';
```

2.4 Creer la Publication (Publieur)

```
USE @PublicationDb;
EXEC sp_replicationdboption @dbname = @PublicationDb, @optname = N'publish', @value =
    N'true';

EXEC sp_addpublication
```

```

@publication = @PublicationName,
@status = N'active',
@allow_push = N'true',
@allow_pull = N'false',
@allow_anonymous = N'false',
@enabled_for_internet = N'false',
@snapshot_in_defaultfolder = N'false',
@alt_snapshot_folder = @SnapshotDir,
@independent_agent = N'true',
@immediate_sync = N'false',
@retention = 72,
@repl_freq = N'continuous',
@sync_method = N'concurrent',
@allow_initialize_from_backup = N'false';

-- Article principal : adapter @ArticleTable
EXEC sp_addpublication_snapshot
    @publication = @PublicationName,
    @frequency_type = 1,
    @frequency_interval = 1,
    @job_login = @ReplLogin,
    @job_password = @ReplPwd;

EXEC sp_addarticle
    @publication = @PublicationName,
    @article = N'Article_Table',
    @source_object = @ArticleTable,
    @type = N'logbased',
    @pre_creation_cmd = N'drop',
    @schema_option = 0x000000000803509F;

```

2.5 Créer l'Abonnement push (Publieur vers Abonne)

```

USE @PublicationDb;
EXEC sp_addsubscription
    @publication = @PublicationName,
    @subscriber = @Subscriber,
    @destination_db = @PublicationDb,
    @subscription_type = N'Push',
    @sync_type = N'automatic',
    @article = N'all',
    @update_mode = N'read only',
    @subscriber_type = 0;

EXEC sp_addpushsubscription_agent
    @publication = @PublicationName,
    @subscriber = @Subscriber,
    @subscriber_db = @PublicationDb,
    @job_login = @ReplLogin,
    @job_password = @ReplPwd,
    @subscriber_security_mode = 0,
    @subscriber_login = @ReplLogin,
    @subscriber_password = @ReplPwd,
    @frequency_type = 64,
    @frequency_interval = 1,
    @frequency_subday = 4,
    @frequency_subday_interval = 1,

```

```
@active_start_time_of_day = 0,  
@enabled_for_syncmgr = N'false',  
@dts_package_location = N'Distributor';
```

2.6 Generer le snapshot initial (Publieur)

```
EXEC sp_startpublication_snapshot @publication = @PublicationName;
```

2.7 Suivre les jobs

```
EXEC msdb.dbo.sp_help_job @job_name = @PublicationName + N'-Snapshot';  
EXEC msdb.dbo.sp_help_job @job_name = @PublicationName + N'-' + @Subscriber + N'-' +  
    @PublicationDb + N'-' + N'1';
```

3 Controles rapides

- Services : `systemctl status mssql-server mssql-server-agent`
- Ports : `ss -lntp | grep 1433`
- UFW : `sudo ufw status`
- Connexion : `sqlcmd -S publieur -U repluser-P'TonMDPSeurise!23'-Q"SELECT@@SERVERNAM
verifiermsdb..MSreplicationagentsetMSdistributionhistory.`

4 Rappel essentiel

L'agent SQL Server doit tourner sur Publieur et Abonne. Sans mssql-server-agent actif, la replication echouera.

5 Besoin d'aide sur /etc/hosts ?

Fournis-moi les IPs et noms (publieur, abonne) et je te donne le bloc pret a coller.